Heterogeneous Causal Effects of In-Hospital Diabetes Medication on 30-Day Readmission

A Double Debiased Machine Learning Approach

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ENSAE Paris

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Introduction

Why is this question important?

- Diabetes costs the U.S. **$327 billion/year** (2017) : 1 in $4 of all healthcare spending
- **7th leading cause of death** in the U.S. : yet outcomes are **highly manageable**
- Key cost driver: **30-day readmissions** : expensive, frequent, and partially preventable

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Research Question

Does inpatient insulin prescription **causally reduce** 30-day readmission, and does the effect **vary across patients**?

The core challenge

Insulin is prescribed to the **sickest** patients $\Rightarrow$ assignment is **not random** $\Rightarrow$ naive regression is biased

Peterson-KFF (2019); ADA, *Diabetes Care* (2018)

Introduction: Project Overview

PICO Framework

P	Diabetic inpatients, 130 US hospitals, 1999–2008
I	Any in-hospital insulin prescription
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Solution: Double Machine Learning (DML)

Regress *residuals* of Y and D (predicted via ML from X) to remove confounding nonparametrically.

The Therapeutic Ladder & Identification Problem

Type 2 diabetes (∼95% of cases)

Metformin (1st line)
effective, cheap, cardioprotective

Add oral agents
sulfonylureas, glitazones. . .

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- Insulin is a **last resort** (reserved for the most severe cases)[1]
- These patients would be readmitted more **regardless** of insulin

[1] HAS, *Therapeutic strategy for the patient living with type 2 diabetes*, May 2024.

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Confounding by indication

Sicker patients get insulin *and* get readmitted more

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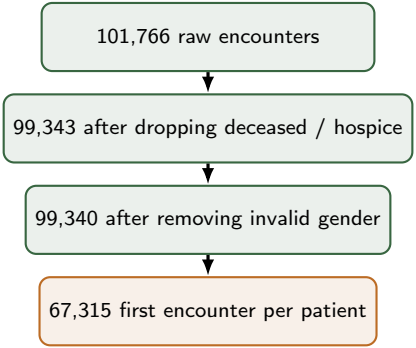
Data analysis

Sample Construction and Key Facts



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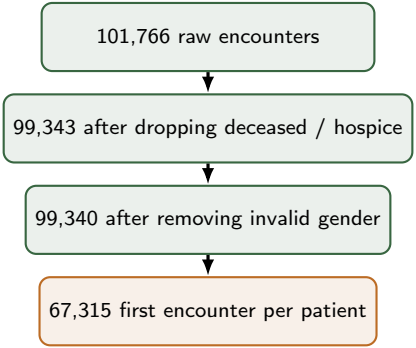

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One row = one patient. This preserves independence and avoids survival-driven selection from repeat admissions.

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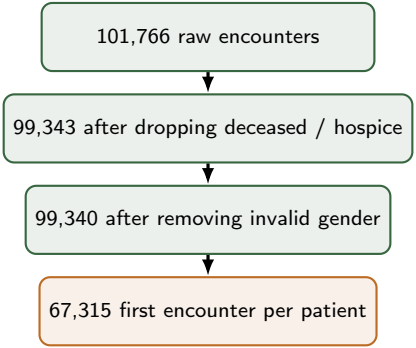

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For identification

- Y: 30-day readmission
- D: any in-hospital insulin
- X: demographics, diagnoses, prior use, labs, medications

Mean age	65.5
Mean LOS	4.25 days
Meds / stay	15.7
Readmission	9.0%
Any insulin	52.1%

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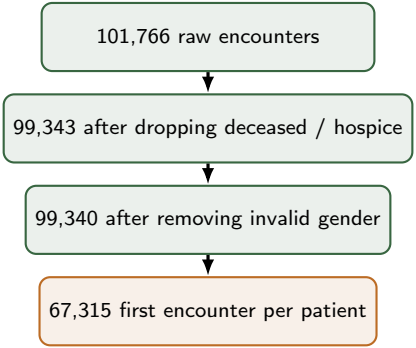

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Missingness is informative: `weight` is 96.9% missing and `A1Cresult` is 83.3% missing, so absence of measurement proxies clinical attention and severity.

Raw Readmission Gradient

Raw pattern: more intensive diabetes treatment is associated with higher 30-day readmission.

Medication regime	N	Share	Readmission rate
No medication	15,730	23.4%	7.65%
Oral only	16,522	24.5%	8.95%
Insulin only	18,944	28.1%	**9.91%**
Insulin + oral	16,119	23.9%	9.40%

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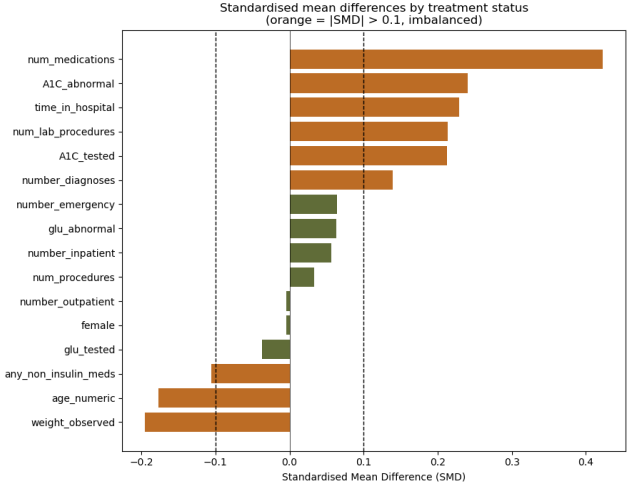
Key project motivation

Naive comparisons make insulin look worse because treatment is targeted to higher-risk patients.

Confounding by Indication



Largest imbalances

Variable	SMD
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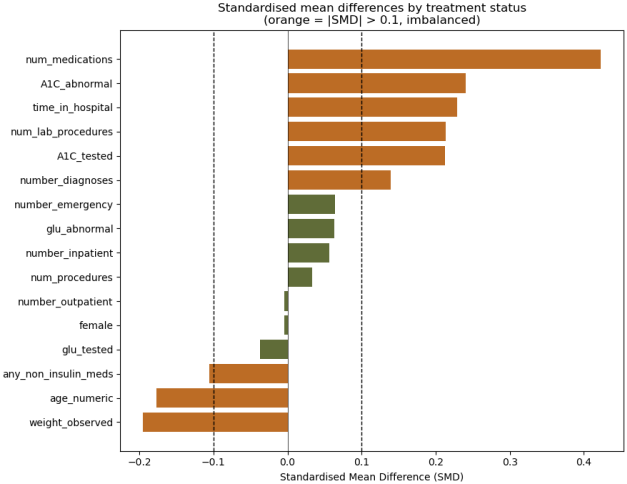

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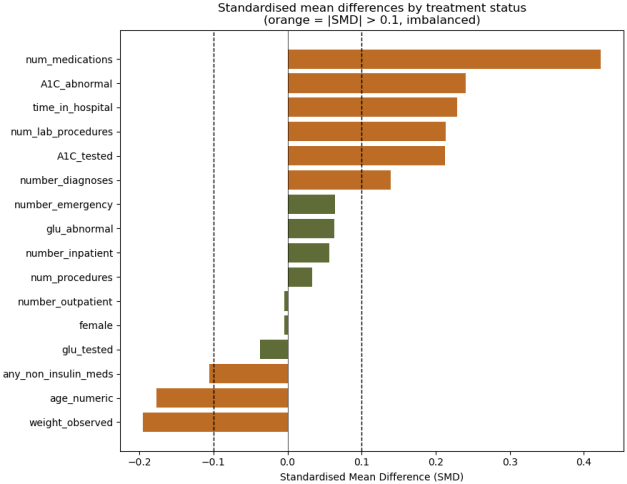

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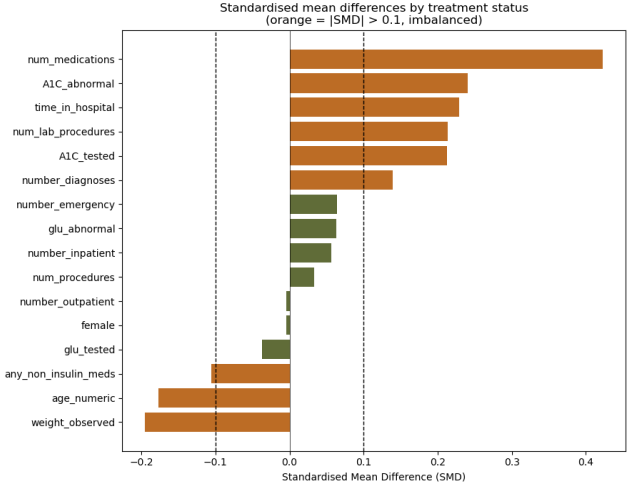

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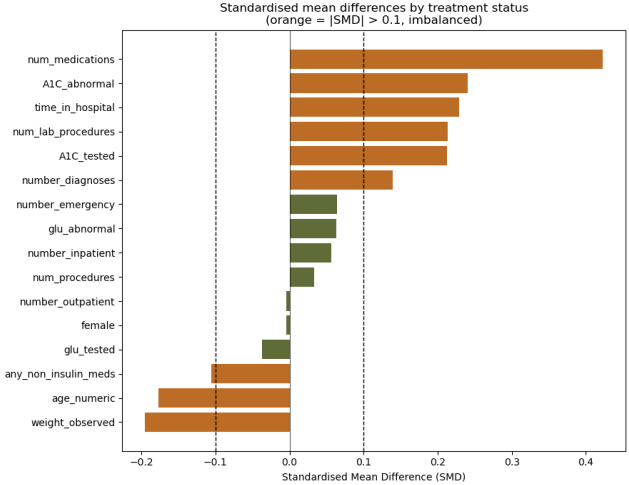

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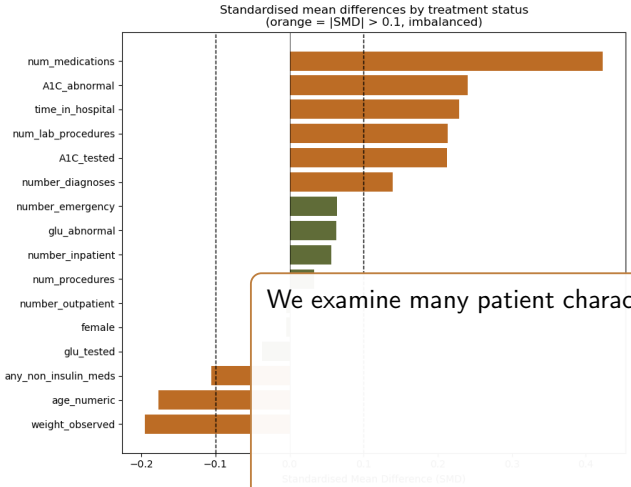

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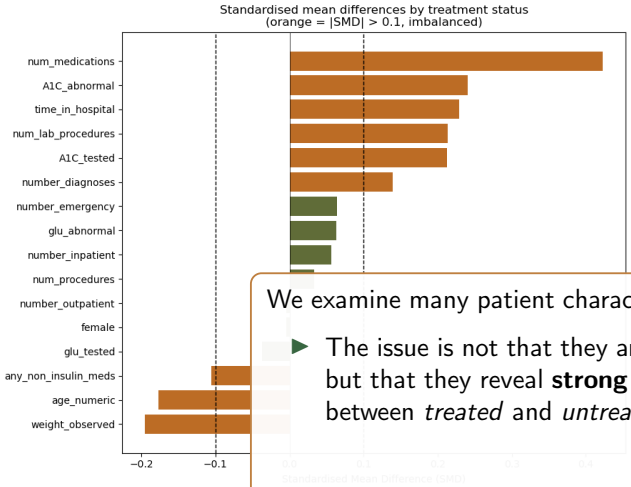

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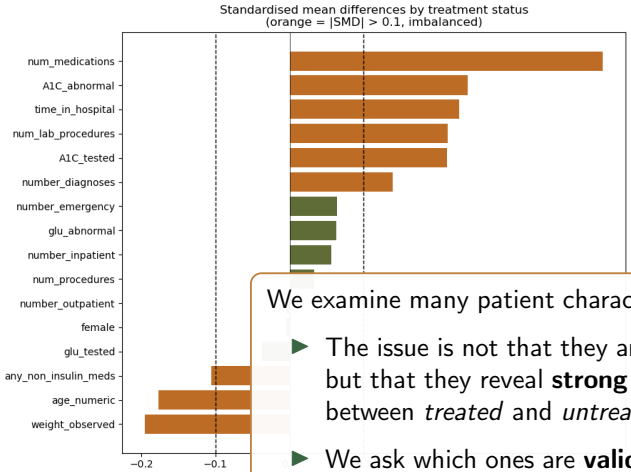

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We examine many patient characteristics here

- The issue is not that they are unusable, but that they reveal **strong imbalance** between *treated* and *untreated* patients
- We ask which ones are **valid pre-treatment controls**

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Identification and Causal Graph

From Variables to a Valid Adjustment Set

Due to timing ambiguity, we classify variables as:

- **Pre-treatment:** admissible for the main adjustment set
- **Time-ambiguous:** potentially reflect the clinical response to insulin prescribing, and could bias the causal pathway if we control for them
- **Excluded / bad controls:** treatment variants or variables too close to the treatment decision

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For the principal X set we retain only the pre-treatment covariates $\rightarrow$ shown in DAG.
This gives **36 variables**.

Causal DAG and the Backdoor Problem

Core identification logic

The main threat is **confounding by severity**: sicker patients who are more likely to receive insulin and are also more likely to be readmitted.

$$D \leftarrow \text{Severity} \rightarrow Y$$

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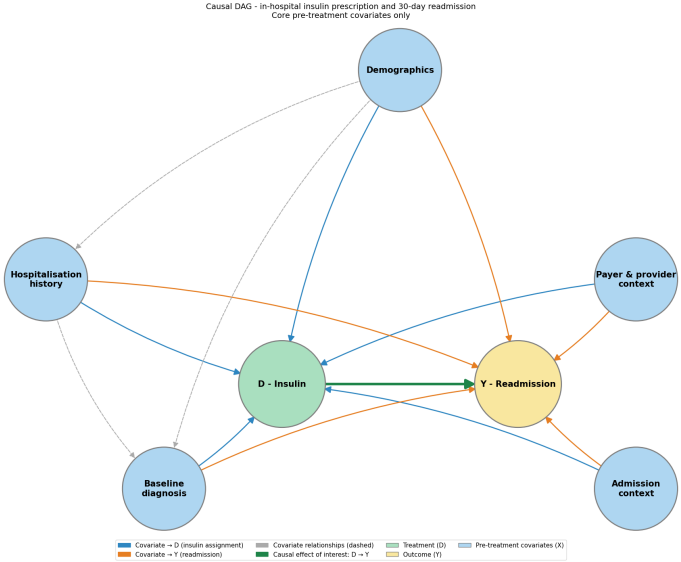
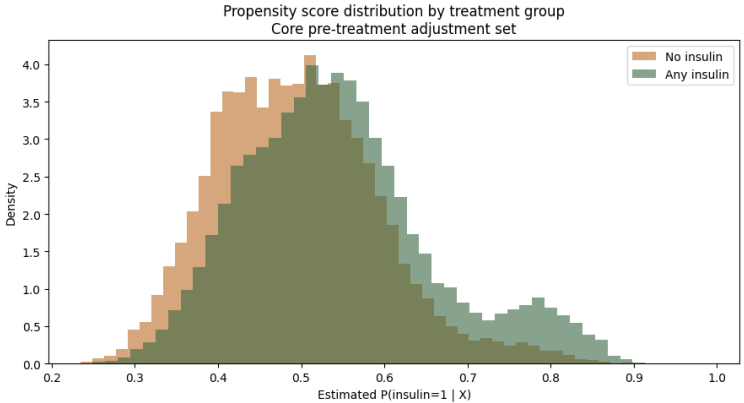
- Because latent severity is unobserved, we block the backdoor path *indirectly* using observed pre-treatment variables in X.

DAG


Causal DAG - in-hospital insulin prescription and 30-day readmission
Core pre-treatment covariates only
Demographics
Hospitalisation history
Payer & provider context
D - Insulin
Y - Readmission
Baseline diagnosis
Admission context
Covariate → D (insulin assignment)
Covariate → Y (readmission)
Covariate relationships (dashed)
Causal effect of interest: D → Y
Treatment (D)
Outcome (Y)
Pre-treatment covariates (X)

Propensity Score – Overlap Check



Diagnostics

AUC	**0.621**

Identification Assumptions and Empirical Plausibility

- **Assumption 1, Unconfoundedness:** Conditional on the core pre-treatment covariates, insulin assignment is treated to be *as close to random as possible.*
  - **Empirical support:** propensity-score distributions of treated and untreated overlap well
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- **Assumption 2, Overlap:** every patient must have a positive probability of receiving or not receiving insulin.
- **Assumption 3, SUTVA:** no interference across patients (highly plausible), and treatment is well-defined for the main analysis.

Statistical estimator/model selection

What estimators do we compare?

Goal: Estimate the causal effect of treatment D on outcome Y

- **Naive LPM**: $Y \sim D \rightarrow$ to study raw association only
- **Adjusted LPM**: $Y \sim D + X \rightarrow$ removes bias from observed confounders

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- **DML-PLR (Partial Linear Regression)**: a more robust model using pre-selected nuisance functions
- **DML-IRM (Interactive Regression Model), our estimator of choice**: designed for binary treatment $\rightarrow$ allows for treatment effect heterogeneity + no strict functional form $\rightarrow$ *ATE and ATT*

The DML pipeline in our notebook

- **Step 1. Define the causal sample:** Y, D, X

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- **Step 5. Retrieve best learner, with smallest RMSE:**
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- **Step 6. Estimate causal effect using best learners:** plug into PLR and IRM to retrieve ATE and ATT

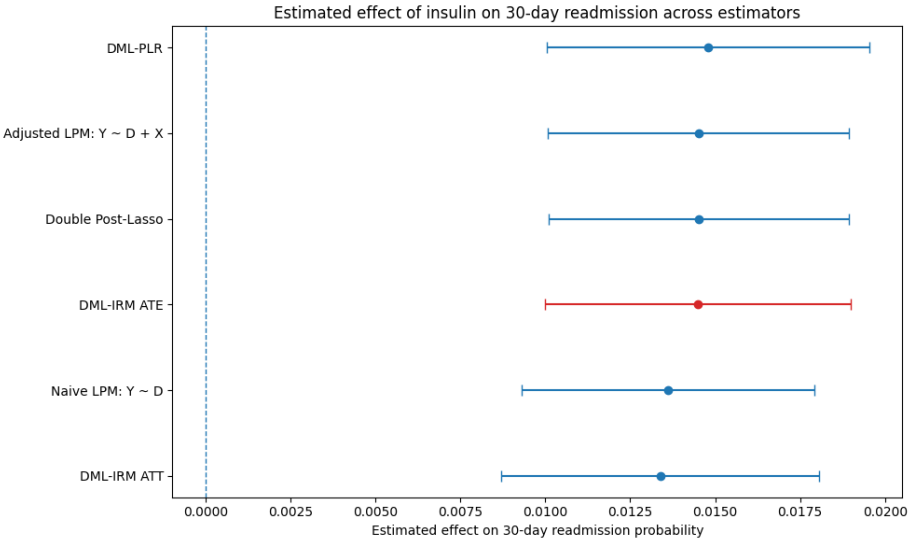
Results: why are the estimates so similar?

Table: Estimated treatment effects across models

Estimator	Estimate (pp.)	Std. Error	p-value
Naive LPM: $Y \sim D$	1.36	0.22	6.49×10^{-10}
Adjusted LPM: $Y \sim D + X$	1.45	0.23	1.32×10^{-10}
Double Post-Lasso	1.45	0.23	1.27×10^{-10}
DML-PLR	1.48	0.24	1.06×10^{-9}
DML-IRM ATE	1.45	0.23	2.84×10^{-10}
DML-IRM ATT	1.34	0.24	2.17×10^{-8}

- Results are robust across specifications. $\rightarrow$ sophisticated IRM confirms this
- Main confounding structure may be close to linear.
- TE is small.

Model comparison as a figure



Heterogeneity + Sensitivity

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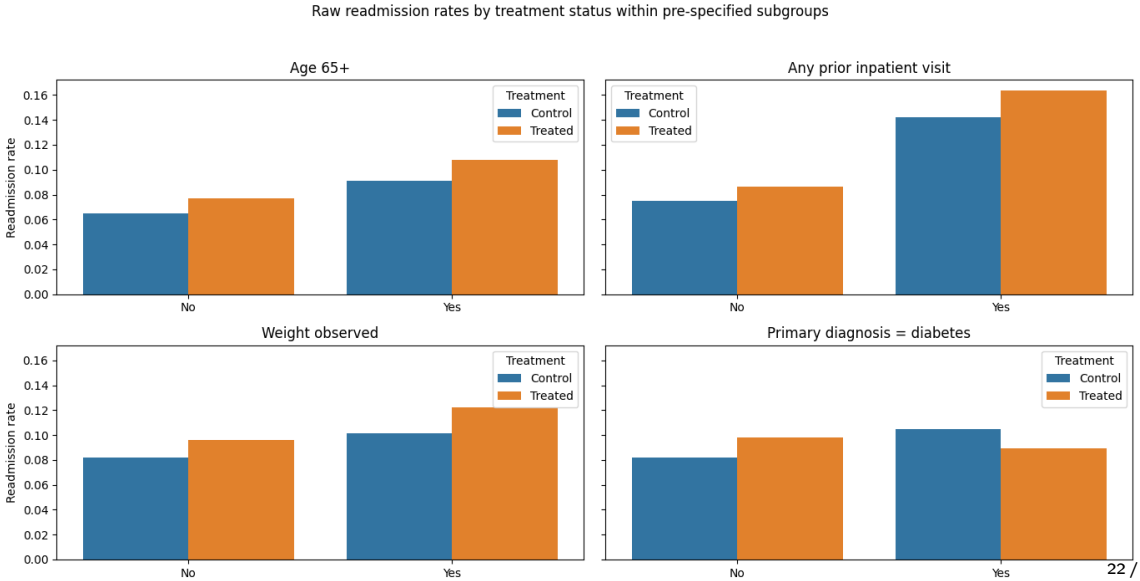
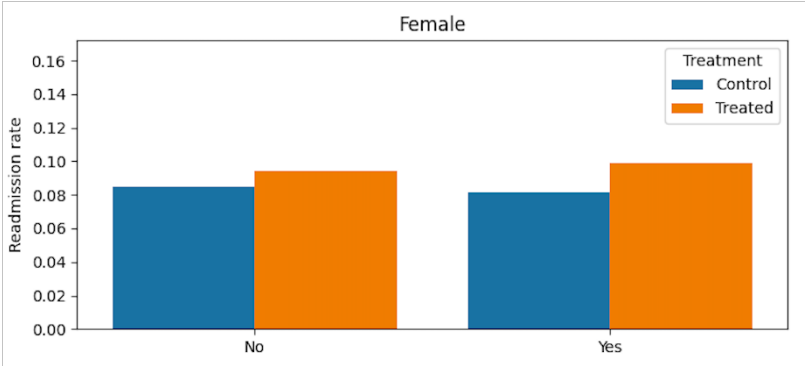


Some descriptives



Heterogeneity: Does the Effect Vary Across Patient Groups?

Table: Heterogeneous treatment effects by subgroup (DML-IRM)

Subgroup	Group	ATE (pp.)	Std. Err.	p-value
Age $\geq$ 65	No	1.10	0.40	0.003
	Yes	**1.70**	0.30	$<$ 0.001
Prior inpatient visit	No	1.40	0.20	$<$ 0.001
	Yes	**1.80**	0.90	0.054
Weight observed	No	1.50	0.20	$<$ 0.001
	Yes	**1.10**	1.90	0.585
Primary diagnosis = diabetes	No	1.70	0.20	$<$ 0.001
	Yes	**-0.40**	1.30	0.766
Female	No	1.10	0.30	0.001
	Yes	**1.80**	0.30	$<$ 0.001

- Larger effect for older patients, women & patients with prior inpatient visits
- Unclear for primary diabetes diagnosis & weight observed $\rightarrow$ small number of observations

Sensitivity Analysis: Alternative treatment definitions

- **Any insulin, our baseline** $\rightarrow$ did the patient receive any insulin at any point?
- **Steady insulin** $\rightarrow$ insulin was given but the dose was not changed
- **Insulin dose increased** $\rightarrow$ the treatment was intensified during the stay
- **Metformin any** $\rightarrow$ we wanted to check how the most popular oral drug behaves as a treatment ($\approx$ 20% of the sample)

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Treatment	Impact on ATE (in pp)	Significance
Any insulin	**+1.45**	***
Steady insulin only	**+0.27**	n.s.
Dose increased	**+1.39**	**
Metformin any	**-0.38**	n.s.

Sensitivity Analysis: Extended covariates set

The extended **covariates set** includes timing-ambiguous variables:

- severity indicators e.g. number of lab procedures done
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DML-IRM: **+0.78 pp**, but still positive and significant (vs +1.45 pp in the conservative X set)

Sensitivity Analysis: Placebo test

Placebo check:

- permuted treatment $\rightarrow$ randomly changing insuline treatment between individuals
- passed: much smaller est., CI includes 0
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Summary

+ **sign** of the effect is robust to different specifications
- **magnitude** is sensitive to treatment definition & to timing-ambiguous covariates

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- **Robust across specifications**: OLS, post-lasso, PLR and IRM all point to the same conclusion
- **Stable in sign, variable in magnitude**: sensitivity and heterogeneity checks

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Limitations

- **Residual confounding**: unobserved severity or physician judgment
- **Temporal ambiguity**: variable measurement is not precise enough to fully separate cause from response
- **Causality remains cautious**: the effect should be interpreted as suggestive rather than fully causal

Quick Quiz

Question 1

Why can we **not** simply compare readmission rates between insulin and non-insulin patients?

A. The sample size is too small
B. Doctors prescribe insulin to sicker patients who are already at higher readmission risk
C. Insulin patients are always older
D. The outcome variable is continuous

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Question 2

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- Some variables are time-ambiguous or too close to the treatment decision, so controlling for them could block part of the causal pathway or introduce bias.
- A good predictor is not automatically a valid control.

Question 3

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- They suggest the sign of the effect is fairly robust, but the magnitude is sensitive to how treatment is defined and which controls are included.

Question 4

After applying DML-IRM, the estimated effect of insulin on 30-day readmission is approximately:

- **A** $+1.45$ pp (insulin slightly increases readmission risk)
- **B** ≈ 0 (no effect at all)
- **C** $+0.07$ pp (small but significant positive effect)
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- A Patients aged ≥ 65
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